Computer Communications 154 (2020) 501–508

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

FCO — Fuzzy constraints applied Cluster Optimization technique for Wireless AdHoc Networks

Amin Salih Mohammed [a], Saravana Balaji B [b], Saleem Basha M S [c], Asha P N [d], Venkatachalam K [e,*]

[a] Department of Computer Engineering, Lebanese French University, Erbil, KR, Iraq
[b] Department of Information Technology, Lebanese French University, Erbil, KR, Iraq
[c] Department of Computing & Informatics, Mazoon College, Oman
[d] Department of Information Science and Engineering, Sapthagiri College of Engineering, Bangalore, India
[e] School of CSE, VIT University Bhopal, India

ARTICLE INFO

Keywords:
Wireless network
Clustering
Fuzzy optimization
Energy efficiency
AdHoc networks

ABSTRACT

Efficient energy utilization in wireless adhoc networks is always an open issue. To achieve energy efficiency, clustering technique would be an optimal one. In clustering technique using advanced optimization methods will provide expected results. By considering this, in this research work fuzzy constraints based cluster optimization technique is proposed. This technique considers fuzzy parameters namely device energy, device position, device movement/speed and device hop-count for calculating the fitness value of the nodes. Based on the value cluster head selection happens. This technique is enhanced to handle uncertainty conditions like cluster head failure, topology changes and energy depletion of the cluster head. The proposed technique is compared with four existing systems and the comparison results illustrates that the proposed technique is more optimal than compared LEACH, MPO, EDC, TCACWCA.

1. Introduction

Wireless adhoc networks are made of wireless devices deployed in a particular geographical area to sense the environment according to the need. Wireless adhoc networks consist of several mobile sensor nodes, and the movement of nodes cause many changes in the topology, and hence there is no fixed infrastructure. These nodes are communicated through the wireless channel and share the information with the base station. These wireless nodes have constrained battery power, and there are many factors that affect the link retention like noise etc. The characteristics of wireless adhoc networks are stated below [1,2].

(1) Wireless Infrastructure: wireless adhoc networks are not having centralized management or any standard infrastructure. Each mobile node can generate the data and can act as router for information forwarding. The architecture behind this is peer to peer architecture. The decentralized infrastructure leads to issues like fault detection and isolation.

(2) Ad hoc Routing: due to wireless infrastructure, there is no standard node for routing. Each node will plays the role of router and forwards the incoming packets towards the destination, if the sending node is not in the radio communication range of the destination node.

(3) Vibrant Topology: As the wireless nodes are regularly moving, it is not possible to have a standard topology structure. The topology of the network changes regularly which leads to node crowd at particular area, rigorous packet loss etc.

(4) Multi Communication Ability: wireless nodes are equipped with radio communication devices. The devices can be extended with multi radio communication devices to transfer data of variable length and communication range. Indeed, this leads to different hardware and software need inside the mobile nodes to support heterogeneous data and communication. This could also leads to disrupted links and data collision. Certain countermeasures are to be taken to avoid those conflicts. To achieve a conflict free mobile communication, development of various protocols and interfaces are required. Development of such kind of protocols and interfaces is a tedious process as conditions like different data types, power constraints, network load, congestion control, flow control are taken into account.

(5) Power Constraint: All the wireless nodes are running with battery power and are constrained with power factor. Nodes cannot lose more energy for processing data or for forwarding the information. Hence the processing procedure and other operations like, information collection, routing procedure should not use more power. This will affect the network life time.

(6) Scalability: As the infrastructure is wireless based, it is possible to scale the network size to vast area. The requirements for this, to

* Corresponding author.
E-mail addresses: kakshar@gmail.com (A.S. Mohammed), saravanabalaji.b@gmail.com (Saravana Balaji B), m.s.saleembasha@gmail.com (Saleem Basha M S), ashapn0311@gmail.com (Asha P N), venkatme83@gmail.com (Venkatachalam K).

https://doi.org/10.1016/j.comcom.2020.02.079
Received 17 June 2019; Received in revised form 19 February 2020; Accepted 27 February 2020
Available online 29 February 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.

adopt the new mobile nodes, capable to attach them with the network, implementing new security measures to handle the scalability etc. The issues are: assigning dynamic addressing for each node to handle network subdivision issue, managing the configuration of wireless nodes, location management, interoperability among wireless nodes etc.

To implement and operate a wireless adhoc network successfully, few issues are to be addressed. Those issues are given below:

(1) Packet Drop: In the wireless networks, mobile nodes will experience frequent packet drop and loss of information. This is due to occurrence of interventions, movement of receiver nodes, mutual interferences, multipath propagation, noise etc.

(2) Frequent Route Changes: As wireless networks support mobility for devices, the topology of the network keeps on changing. This will result in frequent route changes between source and destination. The wireless devices should adhere to the topology changes and respond accordingly.

(3) Battery Constraints: The wireless devices are powered using battery. Batteries are power limited, which results in depletion of network after some time. The processing, sensing and forwarding capability of wireless nodes are constrained by battery power. They should not consume more battery power to prolong the network lifetime.

(4) Network Partitions: In wireless adhoc networks, mobile nodes frequently change their positions. This will lead to occurrence of partitions in the network. The administration of the network must be designed to self-sustain these kind of redundant network partitions.

(5) QoS Fulfillment: For wireless adhoc networks, fulfilling the quality of service requirements is a difficult task. As the network is supporting mobility and wireless transmission, achieving QoS factors like packet delivery ratio, network lifetime etc., will require development of new standards and procedures.

(6) Security: Wireless networks are always prone to security attacks. Security concerns like access control, information integrity, and encryption are requiring more security standards and procedures. In the meanwhile those standards should be lightweight as the entire network infrastructure is battery powered. Several key management systems for providing security are developed in academia and industrial scenario, still security of wireless networks is an open issue.

With research and development activities at academia and industry, the mobile nodes are able to adhere to the environmental changes and act accordingly. The wireless nodes generally report the information to the base station of the network. The base station is in a fixed position. If the wireless nodes are transmitting the information towards base station by themselves using routing protocol, then there will power drain happens at the mobile nodes which will lead to the minimal network lifetime. To avoid this clustering technique can be used. Clustering means a group of mobile nodes will communicate with each other and select one node as a cluster head. In this technique, the mobile nodes are structured themselves into many groups called as clusters. All the nodes in the cluster will collaborate with each other to select one node among them as Cluster Head (CH) [3]. Then all the mobile nodes will transfer the information to the cluster head, and cluster head will take care of forwarding the information to the base station. The cluster head will also take the responsibility of duplicate avoidance, information fusion etc. the advantages of clustering are data fusion, avoiding topology changes, improved QoS of the network and energy efficient routing between the nodes of the network [4]. By utilizing the clustering technique, several objectives could be achieved like overhead reduction, stable connectivity, better bandwidth usage, reduced topology changes and reduced routing table size. Constraint programming is a generic framework for declarative description and effective solving for large, particularly combinatorial, problems. Not only it is based on a strong theoretical foundation but also it is attracting widespread commercial interest as well, in particular, in areas of modeling heterogeneous optimization and satisfaction problems [5]. In this research work, the adhoc network devices scattered in the region are having the following parameters: device energy, device position, device movement, device hop-count, which are all taken into account as fuzzy constraint values as those values will keep on changing and must be satisfying some constraints for keeping the device active in the network. A constraint that is defined on more than two variables is called a global constraint. Based on these constraints, this paper proposes a fuzzy based cluster optimization technique for mobile adhoc networks, aims to prolong the active time of the network.

2. Related Work

Clustering problem is an active study under academia and industry. There are many solutions proposed for clustering in wireless adhoc networks using different computing techniques like conventional, heuristic etc. This section discusses a few of the existing solutions. An energy-efficient clustering algorithm for fixed wireless sensor networks (EECA-F) is proposed in [6]. It is developed to apply in homogeneous fixed wireless sensor networks. This technique calculates the optimum degree of the wireless sensor nodes, to be selected as cluster head. For all the wireless sensor nodes the relevant degree, distance between the base station and the corresponding node, total energy consumed are considered to estimate the weight. The wireless sensor node with minimum weight among all the nodes will act as cluster head. The authors in [7] proposed a clustering protocol called UMBIC which is, Unequal multi-hop balanced immune clustering protocol. This technique mainly focus on the energy consumption criteria of the wireless nodes. This protocol consists of two mechanisms namely: Unequal Clustering Mechanism (UCM) and Multi Objective Immune Algorithm (MOIA). UCM will divide the wireless network into many clusters of random sizes based on the energy of the nodes along with distance of the nodes from the base station. MOIA is used to optimize the cluster size, which were developed in the UCM phase. In this technique base station plays a major part, it involves in finding out most optimal cluster head which can serve the entire cluster and then the optimal route between the cluster head and the base station. The above mentioned process is called as construction phase, where as in the transmission phase cluster heads will transfer the information towards the base station. The most adapted protocol for clustering is LEACH [8]. This protocol is designed for clustering in the wireless network based on micro sensors. This protocol largely achieves the energy efficient data transmission, enhanced routing supporting scalability and provides optimized media access for the micro sensor nodes. It uses random rotation of the cluster heads for evenly distributing the load to the sensors in the cluster to avoid faster energy depletion over few sensor nodes. It also supports data fusion technique that reduces the amount of information transmitted to the base station through cluster heads. Stable load balanced weighted clustered algorithm (SLWCA) is proposed in [9] for homogeneous network clustering. In this algorithm, each node will compute its own weight based on the parameters like power, mobility and degree of orientation. Then all the nodes will share the weight with each other and the node with the lowest weight will become cluster head for the cluster of wireless nodes. Another technique base on parameters like mobility–energy–degree–distance (MED-BS) is proposed in [10]. In this technique also, weight is calculated. Here each node will find the neighboring nodes and the node which is having the lowest mobility factor will calculate the relevant weight of each and every neighbors. The least weighted node will act as cluster head.

In [11], authors proposed an improved weight-based clustering technique called as TCACWCA. This technique utilizes a transmission range clustering algorithm for identifying the nodes that are not within the range and also to check the transmission link consistency. It also uses two metrics called expected transmission count (ETX) and path encounter rate (PER). Based on the two metrics the technique finds the possible collision links and give weight to the mobile nodes. The highest weigh node will be selected as cluster head. Ant colony optimization based dynamic clustering techniques is proposed in [12] (EDC). The proposed technique has a dynamic mechanism to calculate the values

of heuristic variables like reliability, bandwidth, path length, routing delay etc. fuzzy logic issued to find the probability of forwarding path and hybrid fuzzy logic rules are generated. Based on ant colony optimization technique dynamic clusters are created, and the highest energy containing node nominated as head of the cluster. Here, the uncertainty conditions like cluster head failure are not taken into account. Cluster formation using honey bee optimization algorithm is proposed in [13]. This method treats the issue as dynamic optimization problem and honey bee algorithm is used to find the cluster heads with more energy to act as cluster heads. Cluster heads change for every round which makes this technique not optimal as cluster head formation process itself will deplete the energy of the network. A weight based clustering approach called as Cluster Chain Weight Metrics (CCWM) is proposed in [14]. For cluster head selection three parameters are used: minimum path loss factor, node degree and average energy. The node degree defines the number of nodes served by cluster head which is defined as 20 in the method. This makes the algorithm not optimal where new nodes must move to other clusters for communication which makes the energy depletion.

Fire-fly approach based wireless sensor network clustering of nodes is proposed in [15]. This approach has four phases namely: initialization, fetching, intimidation and polishing where the parameters used are residual energy, cluster size and contextual spatial information. In this method, sensor nodes group among themselves to create a set of initial clusters, and the smaller sensor nodes are distribute themselves among the clusters to balance the cluster size. This method is not optimal as smaller sensor nodes are involve in another round for balancing the cluster size which leads to energy depletion at smaller sensor nodes. A genetic algorithm based improved energy balance method of LEACH called LEACH-TLCH is proposed in [16]. In this method various parameters like node energy, node position etc. are taken into account for performing mutation operation and based on the highest fitness function the cluster heads are selected. Based on hop count of the nodes, a clustering technique called Hybrid Energy Efficient Distributed Clustering (HEED) is proposed in [17]. In this method the residual energy of all the sensor nodes are taken in to account and based on the amount of energy remaining, a group of sensor nodes will be nominated as cluster heads. Then all the other sensor nodes will decide to join the cluster or not by using a parameter called as intra cluster communication cost. This method is multi-hop based clustering. After the initial cluster head selection, the cluster heads are uniformly distributed in the network area. In this technique, the entire network structure is needed to know for uniform distribution of cluster heads. HEED provides a uniform CH distribution across the network and better load balancing. While comparing this algorithm with LEACH, it takes the cluster head using the energy remain of each node, where LEACH selects the cluster head randomly.

Dividing the network area into grid cells and forming clusters based on them is proposed in [18]. The network area is divided into grid cells of three type's namely high, low and empty based on the density of the grid cell. Then these grid cells are merged together to form the clusters as normal and advanced clusters. For each cluster, the cluster head role assigned to node which has the maximum level of energy than other nodes. The disadvantage is uncertainty conditions like cluster head failure are not taken into account. Completely energy based cluster head selection is proposed in [19] called as DDEEC (developed DEEC) algorithm. In this technique all the wireless nodes considered into two categories namely: Normal and Advanced. The advanced nodes are determined based on the energy level they are having. At the initial network deployment advanced nodes shall only nominated to operate as cluster heads. During network operation progress, normal nodes will have the chance of becoming cluster heads when the advanced node energy level depleted. A modified algorithm for clustering based on weight called Forecast Weighted Clustering Algorithm (FCWA) is proposed in [20]. The parameters used in this method are forecast weight, the degree of difference, the speed of node and energy of the sensor node. This technique is not promising as the forecasting method is not properly defined. Probability-based optimal cluster head selection based on Energy Efficient Hierarchical Clustering (EEHC) is proposed in [21]. It extends the data gathering model of EEHC based on hierarchy and probability of each node is determined. The node which has maximum probability is acting as a cluster head. Energy prediction based clustering technique using fuzzy logic is proposed in [22]. This method is based on LEACH model where expected energy consumption on sensor nodes is predicted using a neural network model. The expected residual energy is used as fuzzy descriptor during the cluster head selection process.

To minimize the energy consumption in each node and thus extending the network lifetime, a Power-Efficient and Adaptive Clustering Hierarchy (PEACH) protocol [23] is proposed. This protocol understood the characteristics of the wireless network and reduces over head by forming multilevel clustering. This protocol overhears the nodes about the information, and identifies the source and destination of the packets transmitted by the neighboring nodes of the current node under monitoring. This protocol is applicable for location aware wireless networks, and by overhearing it saves the energy consumption on each node and thus achieves the prolonged network lifetime. Indeed, the overhearing process also needs some energy depletion and it is not appropriate for wireless networks with continuously changing network and cluster head selection operation may need to repeat regularly.

In [24], authors proposed a new IWCA algorithm which share the feature of self-adaptability of WCA and is able to adjust corresponding parameters according to different network characteristics, so it has a stable grouped cluster. This work considers the size of cluster based on allocated bandwidth of the network. This will make the far located few nodes not covered in any clusters. It also not deal with failure of cluster head. In [25], authors proposed a new technique for selection of weights for all the nodes participating in a network is done by tuning the effect of weights in terms of throughput and end to end delay with variation of mobility. In this work, the weights are assigned with starting value of 0.05 and incremental value of 0.05, where the real value of energy and other factors are not considered, which affects the performance of the network and frequent cluster head failures. In [26], authors proposed a clustering algorithm inspired by multi-agent stochastic parallel search technique of particle swarm optimization. The cluster formation is presented by taking particle swarm optimization based multi-objective fitness function. In this work the mobility and stability of the nodes and links are taken into account to find the cluster head. This would lead to informal cluster formation if node velocity is higher and stability of the link is lower. It also not dealing with failure of cluster heads. In [27], authors proposed an improved cluster head selection technique using Hybrid Genetic cuckoo search optimization algorithm and location aware clustering. Here, a time sequence is computed for every node and it is allocated to individual time period for every node to allow them to broadcast. This method used prioritized time sequence for broadcasting where the residual energy is calculated using all the nodes information. This will lead to more information processing during cluster setup time and also leads to more energy depletion at the initial stages of the network operation. In [28], authors proposed a hybrid optimization algorithm, named Chronological-Earth Worm optimization Algorithm (C-EWA), for performing effective clustering and adjusts power and energy parameters using topology management. This method takes factors like mobility, link lifetime, power, and distance between nodes into account to construct the topology of the network as a graph. Then clusters are formed and cluster heads selected. This work has the drawback of analyzing the entire network structure to develop the graph structure which needs involvement and message passing of all the nodes of the network. Hence more energy will be depleted during initialization of clusters and network lifetime will be reduced. To overcome the issues discussed above, a new technique called fuzzy constraints based optimized technique is proposed and discussed in the next section.

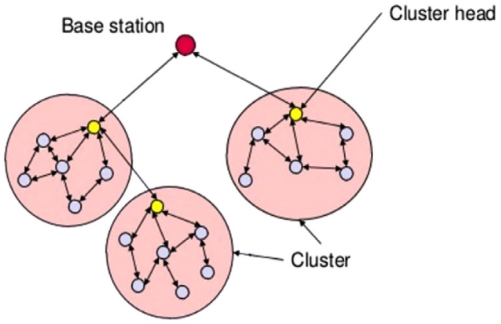


Fig. 1. Wireless adhoc network with clusters.

3. Fuzzy constraints applied optimization technique

During the deployment of the network, the first task is to find the clusters of the network and each cluster with a cluster head. The objectives of the research are 1. To minimize the number of clusters in the network, 2. The number of nodes in the cluster should be the approximately the same quantity in all the clusters so that the energy utilized in the cluster is even throughout the network. The basic structure of a wireless adhoc network after cluster formation is depicted in Fig. 1.

Based on the activity on the network, few mobile nodes will be in the active state, and others are in the sleep state to save the energy utilized in the nodes. After some time the mobile nodes in the sleep state will wake up, and the mobile nodes in the active mode will move to the sleep state. This cycle continues to prolong the network lifetime. As the mobile nodes are moving, or due to the energy depletion in the mobile nodes, the topology of the network changes dynamically from time to time. The main objective of the research work is to discover the cluster head when there is a drastic transformation occurred in the structure of the network. The proposed solution based on fuzzy constraints that depend on the weight allocated for each node and based on it the cluster head is selected when needed.

To find the fitness value f(x) of the node x that is to select as a cluster head, the following parameters taken into account.

(i) *Device energy* (*Ex*): For an optimized selection of cluster head, it is always feasible to select a node with maximum energy than other nodes within the communication range of the particular device. As the cluster head is mostly involved in the transmission of information, cluster heads with lesser energy will drain faster which leads to the selection of cluster head operation again. To avoid, a minimum threshold (50% of total energy) is fixed to avoid the nodes with lesser energy take part in cluster head selection process.

$$Ex = Re/Te \tag{1}$$

where *Te* indicates mobile node total energy and *Re* is indicates mobile node residual energy.

(ii) *Device position* (*Px*): The mobile node that is situated under the radio communication range of the maximum number of mobiles nodes will be optimal for acting as a cluster head. For avoiding the nodes with a lesser number of nodes in communication range, a threshold (30% of a overall nodes) used. Mobile nodes falls above this threshold limit will participate as candidates in the assortment process of cluster head.

$$Px = N/Rx \tag{2}$$

In Eq. (2), *N* indicated the quantity of mobile nodes, Rx is the amount of replies received from nodes.

(iii) *Device movement/speed* (*MSx*): The mobile node with optimal movement direction and speed with other mobiles nodes in the communication range is better for acting as a cluster head. The movement and speed of the mobile node calculated by observing the movement of the other mobile nodes within the communication range.

$$MSx = \vartheta x - \in \vartheta y \tag{3}$$

where ϑx is the movement direction of the mobile node and $\in \vartheta y$ is the relative movement direction of the mobile node with in communication range.

(iv) *Device hop-count* (*Hx*): The mobile node which is in relative position, i.e., one hop distance with all the other nodes in communication range, is an optimal mobile node for acting as a cluster head. For covering the major area of the network, it is considered as the mobile nodes within two-hop distance are neighbors and all the other nodes omitted from the cluster formation.

$$Hx = n(\sum_{1}^{y} y(hc - 1)) \tag{4}$$

where y is the node within communication range and hc is the hop count to reply, the value should be 0 or 1 to accept (i.e., node is within two hop count range). The device energy *(Ex)* plays an important part in keeping the network active and in efficient message dissemination. The device position *(Px)* is important as based on this only, the candidature of the device is decided for cluster membership function and cluster head selection operation. Device movement *(MSx)* is involved in deciding the topology changes of the network and also for adding and removing of devices in the cluster head. Device hop-count *(Hx)* is needed to decide whether the corresponding mobile device is in optimal position to communicate with all the devices of the cluster as well as communicating with other cluster heads it that device is selected as cluster head. Based on the above mentioned criteria's, those four parameters are used for cluster head selection process.

3.1. Setup phase

For setting up the network, it assumed that, mobile nodes are knowing of their current position and their relative position within the network. Based on the coverage area of the network the optimal cluster radius is determined. The radius of the network is denoted as *Rnet* is partitioned into *Z clusters*.

$$Rnet = n(Z) \tag{5}$$

The optimal radius of cluster R*clu* is determined based on Eq. (6).

$$Rclu = Rnet/N(Ex + Px + MSx + Hx) \tag{6}$$

3.2. Formation phase

After completing the setup phase, each node will transmit a neighboring discovery broadcast message within the cluster radius to create its routing table. The routing table of each node will contain information about neighboring nodes, their residual energy and their position in the network. Each mobile node is calculating its fitness value for being a cluster head based on the following fuzzy constraints — Device energy (Ex), Device position (Px), Device movement/speed (MSx) and Device hop-count (Hx). Based on the nodes present in the transmission range using Eq. (4), the node density of each mobile node calculated. The centrality membership function based on Eqs. (7) and (8) of the mobile nodes represents the feasibility of node is central to the surrounding nodes within the communication range. Based on the four fuzzy constraints a trapezoidal membership function is defined for calculating the fuzzy fitness values; i.e. Low, High and Medium fuzzy values of the mobile node.

$$val = \sqrt[Rclu]{\frac{(Ex + Px + MSx + Hx)}{n}} \tag{7}$$

$$mem(x) = \begin{cases} 1 \\ 0 \\ -1 \end{cases} \tag{8}$$

Table 1
Cluster head selection algorithm.

START
Input ∀ (X) position, speed/movement direction (ϑ) and energy (e)
For all mobile nodes (X) calculate

$$Ex = Re/Te$$
$$Px = N/Rx$$
$$MSx = \vartheta x - \in \vartheta y$$
$$Hx = n(\sum_{1}^{y} y(hc-1))$$

Calculate the radius of the cluster

$$Rclu = Rnet\ /\ N(Ex + Px + MSx + Hx)$$

Calculate the number of clusters of the network

$$Rnet = n(Z)$$

Begin cluster formation
For all mobile nodes (X)

$$val = \sqrt[Rclu]{\frac{(Ex + Px + MSx + Hx)}{n}}$$

Calculate the membership value of all node (X)

$$mem(x) = \begin{cases} 1 \\ 0 \\ -1 \end{cases}$$

for all node (X) if mem(X) = 0 | 1
share *val*(x) with ∀ (X) – 1
y = *val*(x) - ∀ (X) {*val*(x)}
if y > 0
act as CH
share node information with ∀ (X) – 1
else
STOP

Each node with mem(x) value 0 or 1 will share its *val* value with the neighboring mobile nodes. Then, each mobile node will compare its *val* value with the values of neighboring nodes received. The node with lesser value will broadcast the values of higher nodes again in the cluster, and the same operation followed until all the nodes are concluded with a single node which has the higher *val* value. Then the corresponding mobile node with higher *val* value will act as the cluster head and transmits its information with all the neighboring mobile nodes. The other nodes that receive this cluster head information will send the request message for joining to the cluster head. Then all the cluster heads in the network will share their corresponding information with the other cluster heads of the network. All the cluster heads will register their information to the base station for the operation of the wireless adhoc network. The algorithm for the cluster head selection is depicted in Table 1.

Uncertainty Handling

To handle uncertainties like cluster head failure, and topology changes, this technique has adapted a fuzzy inference systems (FIS) for the assortment of new cluster head (CH). For monitoring the state of cluster head, each mobile node in the cluster will frequently communicate with echo packets, and cluster head also will communicate with mobile nodes of its cluster to monitor any infrastructure changes. In case of any uncertainty like cluster head failure detected, then the node detected the issue, will initiate the new cluster head identification process by sending the information with other nodes of the cluster. The algorithm for re-cluster head selection is depicted in Table 2.

In the current running phase of the network, each device in the cluster will send periodical beacon messages per every three minutes to the cluster head, to indicate their presence and also to confirm the liveliness of the cluster head. Transmitting beacon messages will consume less energy and hence this process energy consumption is negligible. If any of the devices in not receiving the reply message from the cluster head for three cycles, then the device will decide that the cluster head is inactive and broad cast the information with cluster id to the neighboring nodes. Then step II of algorithm II begins. In this case, the fuzzy constraint defined in Eq. (1) is modified to undertake the new residual energy of the nodes into consideration. Based on the amount of transferred packets from the node, a new energy level will be predicted based on Eq. (9).

$$Ex = \frac{IRe - CRe}{TP} \tag{9}$$

If the value of Ex is greater than the 1, then the candidate node will involve in the new cluster head selection process as per the algorithm I. If the cluster head detects any topology changes or loss of its energy, then it will initiate the new cluster head selection process. This process will also be initiated if there is a topology change in the network. Based on the algorithm II, a new cluster head will be selected under uncertainty conditions, and the network operation continues.

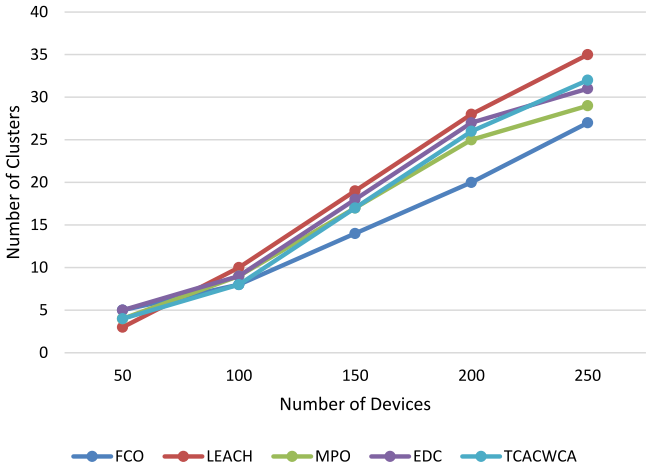


Fig. 2. Comparison on number of clusters.

4. Performance Evaluation

For the evaluation of new techniques and protocols, different simulators are in use. The simulation was implemented based on the network simulator, NS-2 (Version 2.35) [29]. The network region is of size 1000 m * 1000 m and the number of nodes deployed are ranging from 50 to 250. For accessing the medium of the network, CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is used. The two-way ground propagation model is utilized for the simulation of the network. The mobile nodes are able to store the information about their neighbors and have a built-in buffer for the store and forward process. For transmission of packets the methodology adopted is CBR (constant bit rate) and the packet transmission rate is set at 30 packets per second. The simulation is taken for 50 different runs with a different number of nodes and clusters. To evaluate the performance of the proposed technique, it has been compared with existing systems like LEACH [8], MPO [30], EDC [12], and TCACWCA [11]. The metrics used for performance evaluation are no of clusters formed, network lifetime, average delay time and packet delivery ratio. Fig. 2 depicts the performance comparison under a number of clusters formed.

From Fig. 2, it is evident that, the number of cluster established by the proposed fuzzy clustering optimization technique (FCO) is less than the LEACH, MPO, EDC, TCACWCA techniques. If the number of clusters is less, then the number nodes acting as cluster head is lesser, which will need lesser energy for the transmission of data towards base

Table 2
Re-cluster head selection algorithm.

START
Input ∀ (X) position, speed/movement direction (ϑ), initial residual energy (IRe), residual energy (CRe), number of packets transmitted (TP), Cluster Head (CH).
Step I: For all mobile nodes (X)
Begin
For every three minutes
Transmit beacon message to CH
If no reply from CH for three Cycles
CH fails | Topology changes
Goto Step II
End
Step II: Broadcast Information and initiate cluster head selection
Begin
For ∀ (X) calculate new Ex

$$Ex = \frac{IRe - CRe}{TP}$$

Begin Algorithm I.
End
END

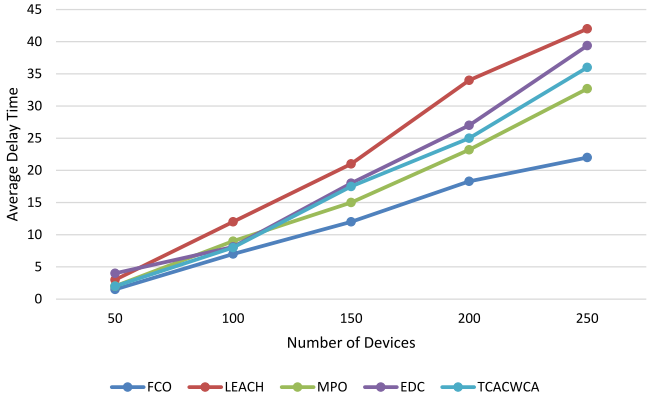


Fig. 3. Comparison on average delay time.

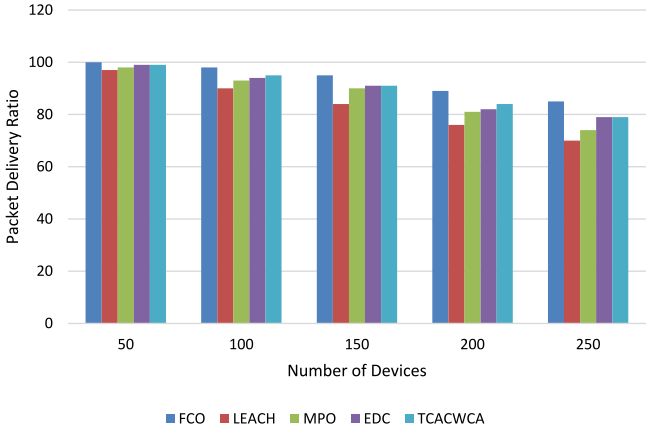


Fig. 4. Comparison of packet delivery ratio.

station. As the optimal hop count for cluster member is only two, all the mobiles nodes are able to transfer the information to the cluster head with minimal energy and the routing also needs minimal energy. This will result in prolonged network lifetime. The performance evaluation in terms of average delay time is presented in Fig. 3.

The Fig. 3 proves that the average delay time of the network is increasing gradually, while LEACH, MPO, EDC, TCACWCA techniques delay time are abrupt. As the maximum hop count inside the cluster is only two, the time taken for the information to reach the cluster head is minimal. Meanwhile the cluster heads able to communicate with the base station in minimal time as the cluster heads are places in a centroid position among other nodes. As minimal delay time is achieved, is inversely proportional to the lifetime of the network. The performance comparison of the proposed technique with the existing systems in terms of packet delivery ratio is provided in Fig. 4.

The packet delivery ratio graph proves that the proposed FCO is more optimal than LEACH, MPO, EDC, TCACWCA techniques. To achieve the objective of prolonged network lifetime, one cannot compromise on packet drop rate. Packet drop occurs when congestion occurs in the network i.e. majority of packet flow occurs through few nodes. As this technique adopts CSMA-CA (Carrier Sense Multiple Access — Collision Avoidance), the network by default tries to avoid collision at nodes and thus improves packet delivery ratio. Meanwhile this technique adopts uncertainty handling methods, will detect the uncertainties of the network which leads to selection of another cluster head and the network operation continues. The performance evaluation in terms of a network lifetime is given in Fig. 5.

Form Fig. 5, it is evident that, the proposed technique is prolonging he network life time than LEACH, MPO, EDC, TCACWCA techniques. The performance of the propose technique in terms of throughput vs. number of iterations is given in Fig. 6.

From the Fig. 6, it is proved that the proposed FCO technique is providing optimal throughput even though number of iterations are increasing while comparing with LEACH, MPO, EDC, TCACWCA techniques. This is because the proposed technique is able to change the cluster head by handling uncertainty conditions. From the above performance evaluation in different terms, it is evident that fuzzy constraints applied clustering optimization techniques is performing well than other existing systems and the objective of prolonging network lifespan is achieved.

5. Conclusion

Clustering technique is the optimal solution for achieving the objective of prolonging lifespan of wireless adhoc networks. There has

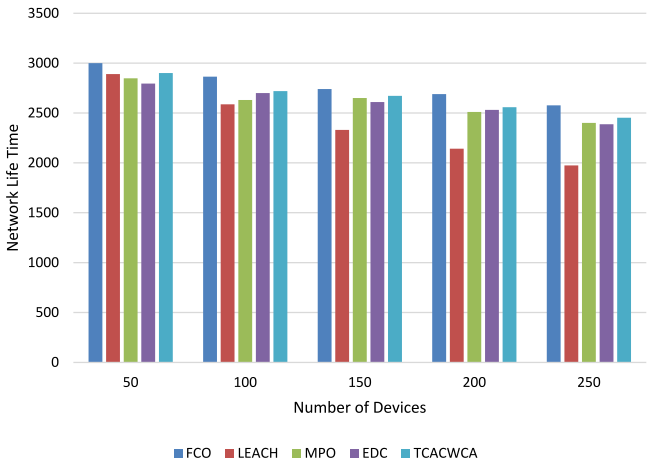


Fig. 5. Comparison of network life time.

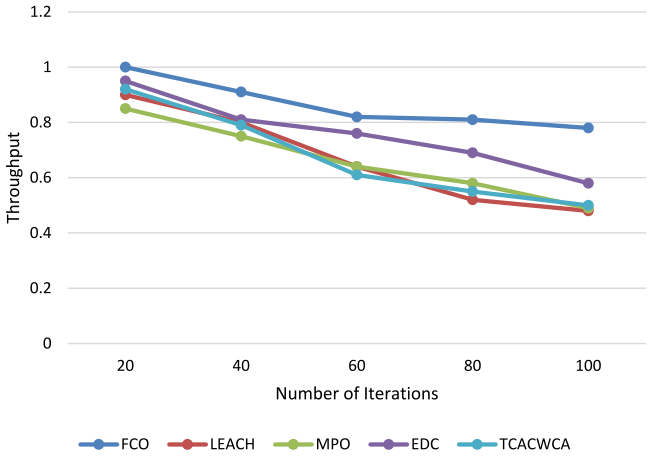


Fig. 6. Comparison of throughput vs. iterations.

been many techniques and protocols are proposed for clustering in wireless adhoc networks, and they have their own drawbacks mainly in handling the uncertainties like cluster head failure, energy depletion in cluster head, topology changes etc. uncertainty handling is well achieved by using different optimization technique. To overcome this kind of issues, in this research fuzzy constraints applied clustering optimization technique is proposed by considering uncertainty handling. Here the constraints of devices of mobile adhoc network like device energy, device position, device movement/speed and device hop-count are taken into account as fuzzy constraints as those values decide the active state of a device in the network. Using those fuzzy constraints, the fuzzy fitness value of each mobile device node is calculated. The mobile device node with the maximum value will act as a cluster head for those node within communication range of a maximum two hop count. In the case of uncertainties, the node detected the issue would initiate the cluster head selection again by taking the values of remaining residual energy of the device and total quantity of the information packets transmitted by that mobile node. The efficiency evaluation of the proposed technique with few other existing systems proves that the fuzzy constraints applied clustering optimization technique achieves the objective of extending network lifetime by handling the uncertainties.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.comcom.2020.02.079.

References

[1] I. Chlamtac, M. Conti, J.J.-N. Liu, Mobile ad hoc networking: imperatives and challenges, Ad Hoc Netw. 1 (1) (2003).
[2] S. Sesay, Z. Yang, J. He, A survey on Mobile Ad Hoc Wireless network, Inf. Technol. J. 3 (2004) 168–175.
[3] W.R. Heinzelman, A. Chandrakasan, H. Balakrishnan, Energy-efficient communication protocol for wireless microsensor networks, in: Proc. 33rd Annu. Hawaii Int. Conf. Syst. Sci, 2000, pp. 3005–3014, http://dx.doi.org/10.1109/HICSS.2000.926982.
[4] M. Ye, C. Li, G. Chen, J. Wu, An energy efficient clustering scheme in wireless sensor networks, Ad Hoc Sens. Wirel. Netw. 1 (2006) 1–21.
[5] A. Algergawy, E. Schallehn, G. Saake, Fuzzy constraint-based schema matching formulation, Scalable Comput.: Pract. Exp. 9 (4) (2008) 303–314.
[6] A. Ben Fradj, N. Nasri, A. Kachouri, Energy-efficient clustering algorithms for fixed and mobile wireless sensor networks, in: Proceedings of the 10th international wireless communications and mobile computing conference IWCMC, Nicosia, Cyprus, 2014.
[7] N. Sabor, M. Abo-Zahhad, S. Sasaki, S. Ahmed, An unequal multi-hop balanced immune clustering protocol for wireless sensor networks, Appl. Soft Comput. 43 (2016) 372–389.
[8] W.R. Heinzelman, A. Chandrakasan, H. Balakrishnan, Energy-efficient communication protocol for wireless microsensor networks, in: Proceedings of the 33rd Annual Hawaii International Conference on System Sciences, Maui, HI, USA, 2000, 10 pp. 2. http://dx.doi.org/10.1109/HICSS.2000.926982.
[9] M. Aissa, A. Belghith, K. Drira, New strategies and extensions in weighted clustering algorithms for mobile ad hoc networks, Procedia Comput. Sci. 19 (2013) 297–304.
[10] A. Ben Fradj, N. Nasri, A. Kachouri, MED-BS clustering algorithm for the small-scale wireless sensors networks, Sci. Res. J. Wirel. Sens. Netw. (WSN) 5 (4) (2013) 67–75.
[11] T. Maragatham, S. Karthik, R.M. Bhavadharini, TCACWCA: transmission and collusion aware clustering with enhanced weight clustering algorithm for mobile ad hoc networks, Cluster Comput. (2018) http://dx.doi.org/10.1007/s10586-017-1574-0.
[12] J. Sathiamoorthy, B. Ramakrishnan, Energy and delay efficient dynamic cluster formation using hybrid AGA with FACO in EAACK MANETs, Wirel. Netw. 23 (371) (2017) http://dx.doi.org/10.1007/s11276-015-1154-2.
[13] M. Ahmad, A.A. Ikram, I. Wahid, M. Inam, N. Ayub, S. Ali, A bio-inspired clustering scheme in wireless sensor networks: BeeWSN, Procedia Comput. Sci. 130 (2018) 206–213, http://dx.doi.org/10.1016/j.procs.2018.04.031.
[14] S. Mahajan, J. Malhotra, S. Sharma, An energy balanced QoS based cluster head selection strategy for WSN, Egyptian Inf. J. 15 (3) (2014) 189–199, http://dx.doi.org/10.1016/j.eij.2014.09.001.
[15] N. Jabeur, A firefly-inspired micro and macro clustering approach for wireless sensor networks, Procedia Comput. Sci. 98 (2016) 132–139, http://dx.doi.org/10.1016/j.procs.2016.09.021.
[16] A.S. Mohammed, Improvement of an energy balance method of LEACH based on genetic algorithm, Control Navig. Commun. Syst. 2 (2017) 136–142.
[17] O. Younis, S. Fahmy, HEED: A hybrid energy-efficient distributed clustering approach for Ad Hoc Sensor Networks, IEEE Trans. Mob. Comput. 3 (4) (2004) 366–379.
[18] M. Abdullah, H.N. Eldin, T. Al-Moshadak, R. Alshaik, I. Al-Anesi, Density grid-based clustering for wireless sensors networks, Procedia Comput. Sci. 65 (2015) 35–47, http://dx.doi.org/10.1016/j.procs.2015.09.074.
[19] B. Elbhiri, R. Saadane, S. El Fkihi, D. Aboutajdine, Developed distributed energy-efficient clustering (DDEEC) for heterogeneous wireless sensor networks, in: 5th International Symposium on I/V Communications and Mobile Network (ISVC), 2010.
[20] Piyalikar, P. Kar, M.K.D. Barma, Forecast weighted clustering in MANET, Procedia Comput. Sci. 89 (2016) 253–260, http://dx.doi.org/10.1016/j.procs.2016.06.054.
[21] D. Phanish, E.J. Coyle, Application-based optimization of multi-level clustering in Ad Hoc and Sensor Networks, IEEE Trans. Wireless Commun. 16 (7) (2017) 4460–4475, http://dx.doi.org/10.1109/TWC.2017.2699175.
[22] J. Lee, W. Cheng, Fuzzy-logic-based clustering approach for wireless sensor networks using energy predication, IEEE Sens. J. 12 (9) (2012) 2891–2897, http://dx.doi.org/10.1109/JSEN.2012.2204737.
[23] S. Yi, J. Heo, Y. Cho, J. Hong, PEACH: power-efficient and adaptive clustering hierarchy protocol for wireless sensor networks, Comput. Commun. 30 (2007) 2842–2852.
[24] H. Yang, Z. Li, A clustering optimization algorithm based on WCA in MANET, in: 2015 6th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Denton, TX, 2015, pp. 1–5.
[25] K. Prasai, S. Ghimire, Selection of Weighting Factors in Weighted Clustering Algorithm in MANET, World Congress on Engineering and Computer Science 2016 Vol I WCECS 2016, October 2016, 19–21, San Francisco, USA.

[26] N. Khatoon, Amritanjali, Mobility aware energy efficient clustering for MANET: A bio-inspired approach with particle swarm optimization, Wirel. Commun. Mob. Comput. 2017 (2017) 1903190, 12 pages.

[27] J. Banumathi, R. Kanthavel, Node failure aware broadcasting mechanism in mobile adhoc network environment, Program. Comput. Softw. 44 (6) (2018) 371–380.

[28] B. Devika, P.N. Sudha, Power optimization in MANET using topology management, Eng. Sci. Technol. Int. J. (2019).

[29] UCB/LBNL/VINT network simulator: NS-2, 2000, [Online]. Available: http://www.isi.edu/nsnam/ns/.

[30] Su. Suganthi, S.P. Rajagopalan, Multi-swarm particle swarm optimization for energy-effective clustering in wireless sensor networks, Wirel. Pers. Commun. 94 (4) (2017) 2487–2497, http://dx.doi.org/10.1007/s11277-016-3564-6.